



Figure 2: Line Mode browser (Credits: https://en.wikipedia.org/wiki/Line_Mode_Browser)

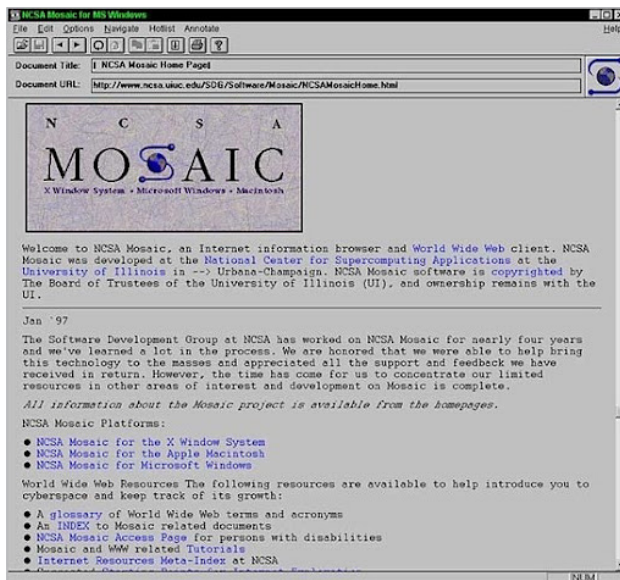


Figure 3: Mosaic Web browser (Credits: http://memex.cc/Web_browser)

Creation of JavaScript

Initially, Web pages were static. In 1995, Netscape decided to add a scripting language to its Navigator. It pursued two routes to achieve this: collaborating with Sun Microsystems to embed the Java programming language, while also hiring Brendan Eich to embed the Schema language. In September 1995, Eich created the first version of JavaScript (officially called LiveScript), which was developed in just ten days. The name was changed from LiveScript to JavaScript in three months, due to marketing reasons, as Java was the programming language trending at that time.

How does a Web browser work?

A Web browser is a client program that allows clients and servers to communicate with each other. When the Web address or uniform resource locator (URL) in your browser is given, the Web browser actually sends an HTTP command to the Web server, directing it to fetch and transmit the requested Web page and display the information needed on the browser.

To display the Web page on the screen of the browser, a group of structured codes (as shown in Figure 4) is needed to

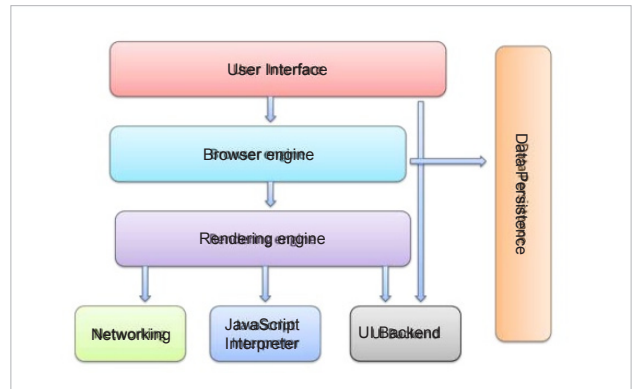


Figure 4: Browser architecture (Credits: <https://medium.com/web-god-mode/how-web-browsers-work-behind-the-scene-architecture-technologies-and-internal-working-fec601488bfa>)

work together and perform a series of tasks.

Let us look at a few aspects of the Web browser.

1. **User interface:** Input given to the browser such as URL, Next, Back, etc.
2. **Browser engine:** Based on the input from the user interface, it queries and manipulates the rendering engine.
3. **Rendering engine:** Interprets the HTML, XML documents and images that are formatted using cascading style sheets (CSS), and generates the layout that is displayed in the user interface.
4. **Networking:** Retrieves the URLs using the common Internet protocols of HTTP or FTP.
5. **JavaScript interpreter:** This interprets and executes the JavaScript code embedded in a website on the browser. The interpreted results are sent to the rendering engine for display. If it's an external script, the resource is first fetched from the network. Parser, the compiler that breaks data into smaller elements, keeps the results on hold until the script is executed.
6. **UI backend:** User interface backend is used for drawing basic widgets like combo boxes and windows. This backend exposes a generic interface that is not platform-specific. It uses operating system user interface methods underneath.
7. **Data persistence/storage:** This is a persistence layer. Browsers support storage mechanisms such as localStorage, IndexedDB, WebSQL and FileSystem. It is a small database created on the local drive of the computer where the browser is installed. It manages user data such as cache, cookies, bookmarks and preferences.

It is important to note that all Web browsers are not created equal; hence, their respective displays will be different.

JavaScript engine

JavaScript is a client side language. It can also be called a Web browser language. HTML and CSS display the content in the Web page, and JavaScript is used for interaction.